

Green's Identity and Method of Images for the Half-Space

*Unofficial personal solution to Cambridge Mathematical Tripos Part IB 2025, Paper 3, Question 14D
(Methods).*

Yucheng (Luca) Jin

Differential Equations

Prove that, for scalar fields $\phi(\mathbf{x})$ and $\psi(\mathbf{x})$ in a three-dimensional domain $\mathcal{D}$ with boundary $\partial\mathcal{D}$,

$$\int_{\mathcal{D}} (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV \equiv \int_{\partial\mathcal{D}} \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) dS, \quad (1)$$

where $\mathbf{n}$ is the outward unit normal to $\partial\mathcal{D}$.

Let $\psi(\mathbf{x})$ satisfy

$$\nabla^2 \psi = \delta(\mathbf{x} - \mathbf{x}_0) \quad \text{in } z > 0, \quad (2)$$

with

$$\psi = 0 \text{ on } z = 0, \quad \psi \rightarrow 0 \text{ as } |\mathbf{x}| \rightarrow \infty, \quad (3)$$

in Cartesian coordinates $\mathbf{x} = (x, y, z)$. Use the method of images to determine $\psi(x, y, z)$.

Now use the identity above to solve the equation

$$\nabla^2 \phi = 0 \quad \text{in } z > 0, \quad \phi \rightarrow 0 \text{ as } |\mathbf{x}| \rightarrow \infty, \quad (4)$$

with $\phi = 1$ on $z = 0$, $x^2 + y^2 < 1$, while $\phi = 0$ on $z = 0$, $x^2 + y^2 > 1$, in terms of a surface integral. Find the closed-form solution for $\phi(0, 0, z)$.

Solution

Write the integrand as a divergence:

$$\phi \nabla^2 \psi - \psi \nabla^2 \phi = \nabla \cdot (\phi \nabla \psi) - \nabla \phi \cdot \nabla \psi - \nabla \cdot (\psi \nabla \phi) + \nabla \psi \cdot \nabla \phi = \nabla \cdot (\phi \nabla \psi - \psi \nabla \phi). \quad (5)$$

Applying the divergence theorem immediately gives the result.

The free-space Green's function satisfying $\nabla^2 G = \delta(\mathbf{x} - \mathbf{x}_0)$ with $G \rightarrow 0$ at infinity is $G = -1/4\pi|\mathbf{x} - \mathbf{x}_0|$. To impose $\psi = 0$ on $z = 0$, place an image of opposite sign at the reflected point $\bar{\mathbf{x}}_0 = (x_0, y_0, -z_0)$:

$$\psi(\mathbf{x}) = -\frac{1}{4\pi} \left(\frac{1}{|\mathbf{x} - \mathbf{x}_0|} - \frac{1}{|\mathbf{x} - \bar{\mathbf{x}}_0|} \right). \quad (6)$$

On $z = 0$, $|\mathbf{x} - \mathbf{x}_0| = |\mathbf{x} - \bar{\mathbf{x}}_0|$ so $\psi = 0$.

Apply Green's identity to ϕ (harmonic) and ψ ($\nabla^2 \psi = \delta(\mathbf{x} - \mathbf{x}_0)$) over $z > 0$:

$$\phi(\mathbf{x}_0) = \int_{\partial\mathcal{D}} \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) dS. \quad (7)$$

The boundary has two parts: the plane $z = 0$ and a sphere at infinity. The sphere at infinity contributes nothing since $\phi, \psi \rightarrow 0$. On $z = 0$, $\psi = 0$ so the $\psi \partial \phi / \partial n$ term vanishes. The outward normal from $z > 0$ at $z = 0$ is $\hat{\mathbf{n}} = -\hat{\mathbf{z}}$, so $\partial \psi / \partial n = -\partial \psi / \partial z|_{z=0}$. Computing (setting $x_0 = y_0 = 0$ for $\phi(0, 0, z_0)$):

$$\frac{\partial \psi}{\partial z} \Big|_{z=0} = -\frac{1}{4\pi} \left[-\frac{z - z_0}{|\mathbf{x} - \mathbf{x}_0|^3} + \frac{z + z_0}{|\mathbf{x} - \bar{\mathbf{x}}_0|^3} \right]_{z=0} = -\frac{z_0}{2\pi(x^2 + y^2 + z_0^2)^{3/2}}, \quad (8)$$

so $\partial \psi / \partial n|_{z=0} = z_0 / 2\pi(x^2 + y^2 + z_0^2)^{3/2} > 0$. Since $\phi = 1$ on $x^2 + y^2 < 1$ and zero otherwise:

$$\phi(0, 0, z_0) = \int_{x^2 + y^2 < 1, z=0} \frac{z_0}{2\pi(x^2 + y^2 + z_0^2)^{3/2}} dS. \quad (9)$$

Converting to polar coordinates $r^2 = x^2 + y^2$:

$$\phi(0, 0, z_0) \quad (10)$$

$$\begin{aligned} &= \int_0^{2\pi} d\theta \int_0^1 \frac{z_0}{2\pi(r^2 + z_0^2)^{3/2}} r dr = z_0 \int_0^1 \frac{r dr}{(r^2 + z_0^2)^{3/2}} \\ &= z_0 \left[-\frac{1}{\sqrt{r^2 + z_0^2}} \right]_0^1 = z_0 \left(\frac{1}{z_0} - \frac{1}{\sqrt{1 + z_0^2}} \right). \end{aligned} \quad (11)$$

Hence:

$$\boxed{\phi(0, 0, z_0) = 1 - \frac{z_0}{\sqrt{1 + z_0^2}}}. \quad (12)$$

As $z_0 \rightarrow 0^+$, $\phi \rightarrow 1$ (matches the BC on the unit disk). As $z_0 \rightarrow \infty$, $\phi \rightarrow 0$ (matches the far-field BC).